INCIDENT RESPONSE SIMULATION REPORT

Windows Event Log Analysis · Task 4 · Alfido Tech Internship

Prepared By	Arathi Shekhar Munavalli	**Date**	05-04-2026
Organization	Alfido Tech Internship	**Task**	Task 4 — Incident Response Simulation
Log Source	Windows Security Event Log	**Total Events**	28,127
Tool Used	Windows Event Viewer (Built-in)	**Period**	05-04-2026, 14:53 – 16:18

1. Introduction

This report documents the findings of an Incident Response Simulation exercise performed as part of the Alfido Tech Cybersecurity Internship (Task 4). The objective was to analyze suspicious activity in the Windows Security Event Log, identify possible Indicators of Compromise (IOCs), and document appropriate response and containment actions. The analysis was performed using the built-in Windows Event Viewer on a Windows 11 machine.

2. Scope & Methodology

Parameter	Details
Log Source	Windows Event Viewer → Windows Logs → Security
Total Events	28,127
Events Analyzed	Event ID 4624 (587 events), Event ID 4672 (558 events), Event ID 4798 (3,085 events)
Analysis Period	05-04-2026, approximately 14:53 to 16:18
Operating System	Windows 11
Tool Used	Windows Event Viewer (built-in)

Methodology — Phases:

- Phase 1: Opened Windows Event Viewer and navigated to Windows Logs → Security
- Phase 2: Applied filters individually for Event IDs 4624, 4672, and 4798
- Phase 3: Recorded event counts, timestamps, and frequency patterns
- Phase 4: Identified anomalies, documented IOCs, and prepared this report

3. Event Log Analysis Results

Event ID	Event Name	Count	Category	Description
4624	Successful Logon	587 (2.1%)	Logon	Account successfully logged on to the system
4672	Special Privileges Assigned	558 (2.0%)	Special Logon	Sensitive privileges assigned to new logon session
4798	Group Membership Enumerated	3,085 (10.9%)	User Acct Mgmt	Process enumerated user's local group memberships

3.1 Event ID 4624 — Successful Account Logon (587 events)

Event 4624 is generated every time an account successfully logs on to the system. A total of 587 such events were recorded during the analysis period. Multiple logon events were observed at 1–2 second intervals (e.g., 15:55:32, 15:55:33, 15:55:35), which is atypical for a normal interactive user session and may indicate scripted or automated logon activity.

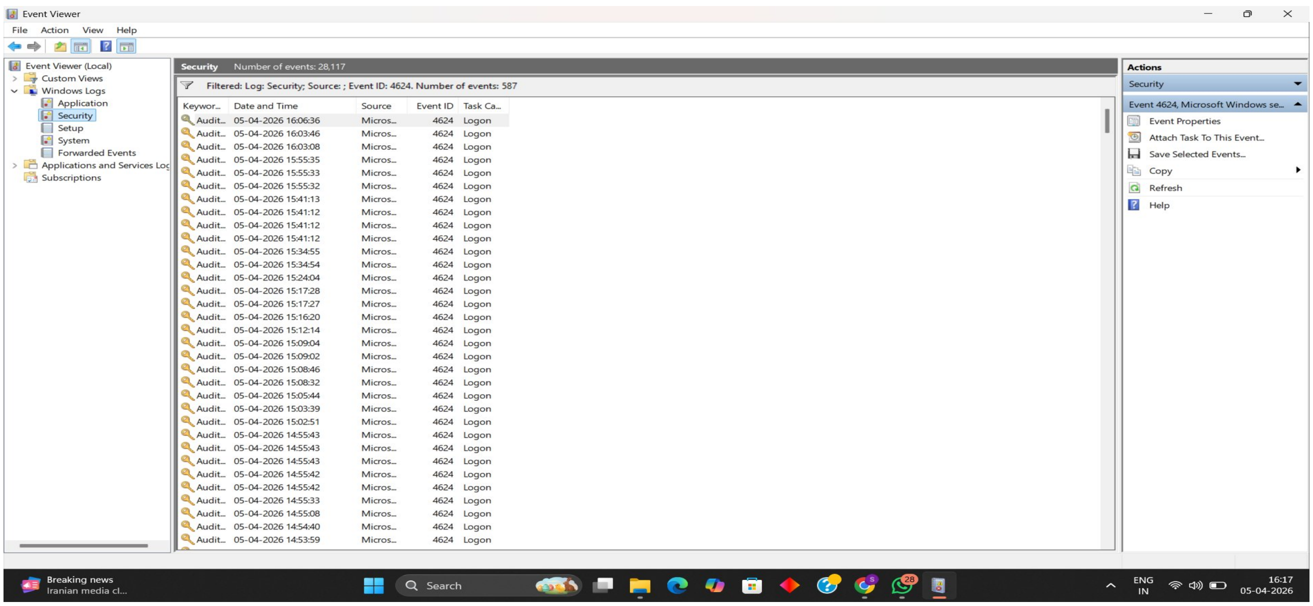

Figure 1 — Event ID 4624 filtered view: 587 Successful Logon events

3.2 Event ID 4672 — Special Privileges Assigned (558 events)

Event 4672 is logged when a new logon session is assigned sensitive privileges such as SeDebugPrivilege, SeTcbPrivilege, or SeBackupPrivilege — privileges associated with administrative or SYSTEM-level access. 558 such events were observed, closely mirroring the 4624 logon timestamps, indicating that nearly every logon session received elevated privileges.

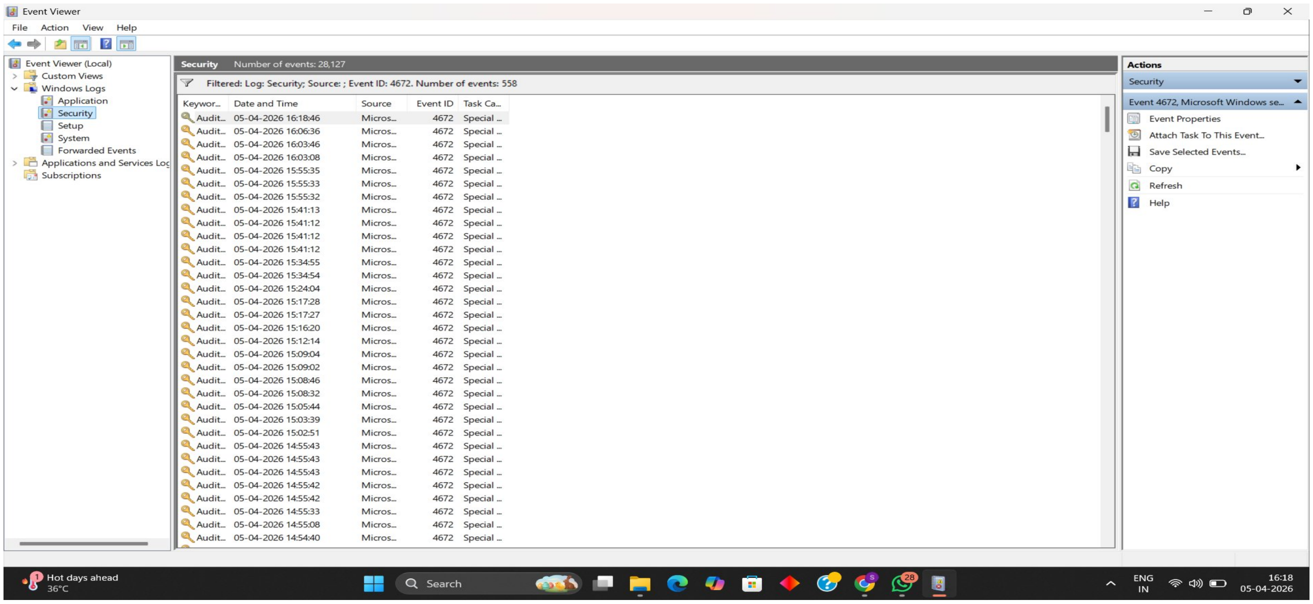

Figure 2 — Event ID 4672 filtered view: 558 Special Logon events

3.3 Event ID 4798 — User Group Membership Enumerated (3,085 events)

Event 4798 is generated when a process enumerates a user's local group memberships. A count of 3,085 is significantly higher than the combined logon events (1,145), representing a 2.7:1 ratio. Burst clusters were observed at 16:10:32, 16:11:44, 16:14:32, and 16:16:32 — with multiple events firing within the same second. This pattern is strongly consistent with automated reconnaissance tools or scripts performing bulk group enumeration.

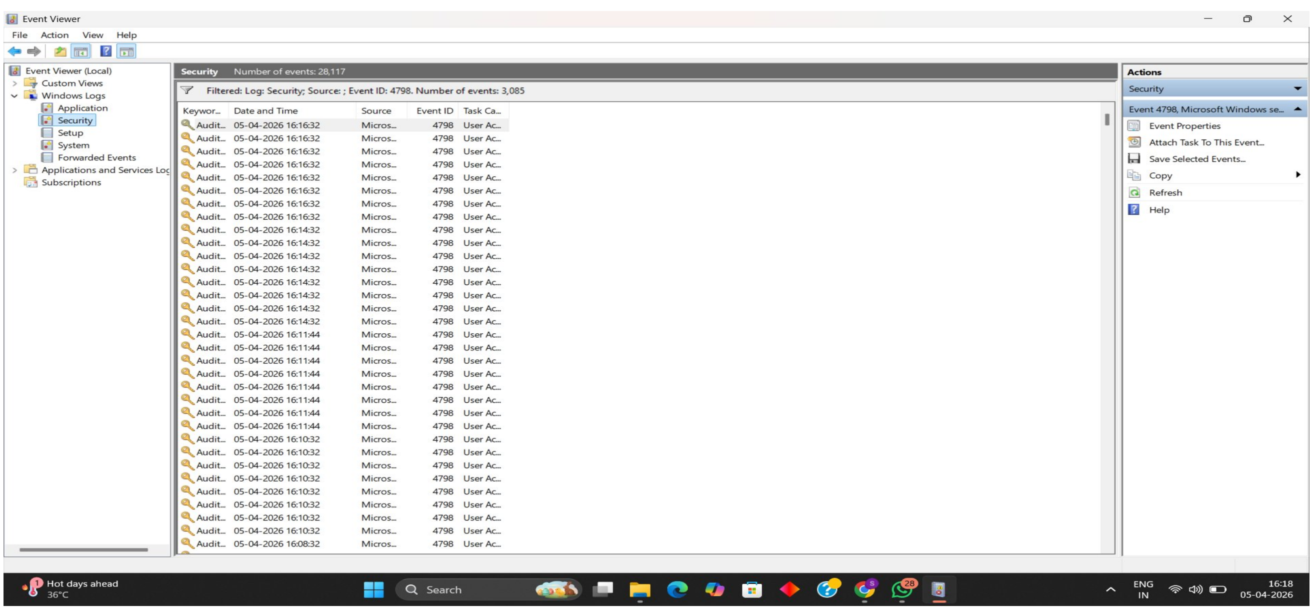

Figure 3 — Event ID 4798 filtered view: 3,085 Group Enumeration events

4. Detected Anomalies

#	Anomaly	Details	Event ID	Severity
1	Rapid Successive Logons	Multiple 4624/4672 events at 1-sec intervals (15:55:32, :33, :35)	4624, 4672	Medium
2	Excessive Group Enumeration	3,085 × 4798 vs 587 logons — 5:1 ratio is abnormal	4798	High
3	Burst Cluster Pattern	Multi-events per second at 16:10, 16:11, 16:14, 16:16 — scripted	4798	High
4	Continuous Privilege Assignment	558 special privilege events closely matching logon volume	4672	Medium

5. Risk Summary

HIGH	MEDIUM	LOW	INFO
2	2	0	0

6. Indicators of Compromise (IOCs)

IOC	Type	Description
3,085 × Event ID 4798	Volume Anomaly	Excessive group enumeration — 5x higher than logon count
Burst at 16:10–16:16	Temporal Pattern	Multiple events per second in tight repeated clusters
558 × Event ID 4672	Privilege Anomaly	Special privileges assigned at every logon — persistent admin
1-sec logon intervals	Behavioral Pattern	Rapid successive 4624 events indicate scripted/automated logon

7. Response & Containment Actions

Immediate Steps Taken:

- Opened Windows Event Viewer and navigated to Windows Logs → Security
- Applied filters for Event IDs 4624, 4672, and 4798 to isolate suspicious activity
- Recorded event counts, burst timestamps, and frequency patterns as documented evidence
- Identified anomalous volume and clustering patterns as potential Indicators of Compromise

Recommended Containment Actions:

- Correlate 4798 burst events with running processes using Process Monitor to identify the source application
- Inspect scheduled tasks and startup scripts for automated logon sequences
- Review all accounts holding SeDebugPrivilege / SeTcbPrivilege — restrict to minimum necessary
- Enable Event ID 4688 (Process Creation) audit policy to trace which process triggered the 4798 spikes
- Deploy Sysmon with a community ruleset for deeper endpoint process and network visibility
- If malicious activity confirmed — isolate endpoint immediately and preserve the .evtx log as forensic evidence
- Configure SIEM alerting rules for bulk 4798 events exceeding 100 occurrences per minute

8. Recommendations

- Enable advanced audit policies including Process Creation (Event 4688) for full process-level visibility.
- Deploy Sysmon with the SwiftOnSecurity ruleset for deeper endpoint telemetry.
- Integrate Windows Security logs into a SIEM platform (Splunk / Elastic) for real-time alerting.
- Regularly review privileged accounts — restrict SeDebugPrivilege and SeTcbPrivilege to necessity only.
- Create automated alert rules for any Event ID 4798 count exceeding 500 occurrences in a single hour.
- Conduct periodic log reviews to proactively detect lateral movement or credential harvesting activity.

9. Conclusion

The Windows Security Event Log analysis conducted on 05-04-2026 revealed significant anomalies across three event categories. While Event IDs 4624 and 4672 volumes are within plausible ranges for an active workstation, the 3,085 occurrences of Event ID 4798 in tight burst clusters represent a strong indicator of automated group enumeration — a technique commonly associated with reconnaissance during lateral movement or credential harvesting attacks.

No direct evidence of a confirmed breach was found through log analysis alone. However, the observed patterns are sufficient to warrant a deeper forensic investigation, including process-level analysis and network log correlation. All findings, IOCs, and recommended response actions have been fully documented in this report.